Supplementary Material: physical anchoring, Lyapunov–Floquet consistency and matched-measurement geometric bound for dissipative optomechanics with synthetic flux

Stella Rolande Mbokop Tchounda[1], Carolle Tchodimou[1], Philippe Djorwe[2], Sifeu Takougang Kingni[3] and Serge Guy Nana Engo[1⋆]

1 Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon
2 Department of Physics, Faculty of Science, University of Ngaoundéré, P.O. Box 454, Ngaoundéré, Cameroon
3 Department of Mechanical, Petroleum and Gas Engineering, National Advanced School of Mines and Petroleum Industries, University of Maroua, P.O. Box 46, Maroua, Cameroon
⋆ serge.nana-engo@facsciences-uy1.cm

September 4, 2026

1 Overview and reading guide

This Supplementary Material supports the main text by providing the normalization, anchor selection, physically anchored reference table, the formal statements of the Lyapunov–Floquet consistency criterion (primary certification of hyperchaos order) and the matched-measurement Fisher lemma (secondary geometric bound, with their proofs), the matched measurement-Fisher reference tables and their observation-window robustness check, the representative finite-time convergence of the strong-coupling spectrum, the threshold sensitivity of the hyperchaos-order classification, the reduced-model parameter screen, the amplitude-quadrature noise-model cross-check against a quantum master equation, the matched measurement Fisher calculation on the chaotic attractor and the semiclassical boundary of the strong-coupling classification, and the explicit technical backup that the Discussion in the main text compresses: a per-gap limitations statement, a per-parameter feasibility audit against the silicon optomechanical anchor, a per-seed robustness decomposition of the long-window trajectory audit, and a phase-sweep and single-mode collapse of the matched Fisher on the chaotic attractor. The reading order that follows the main text is: anchors (Section 2); matched Fisher (Section 4); theorems and proofs (Section 5); convergence (Section 6); threshold sensitivity (Section 7); observation-window check (Section 8); reduced-model screen (Section 9); noise cross-check (Section 10); interpretation (Section 11); attractor Fisher (Section 12); semiclassical boundary (Section 13); limitations and residual gaps (Section 14); feasibility audit (Section 15); initial-condition robustness (Section 16); attractor-Fisher audit (Section 17). Every quantitative claim in the main text is traceable to a table or deterministic script invoked below.

2 Normalization and anchor selection

The main text uses a rotating-frame Hamiltonian normalized to a reference mechanical frequency ω_m, so that all rates and couplings are reported as dimensionless ratios x/ω_m. A physically anchored reference set is obtained by fixing ω_m from a device with gigahertz

mechanical modes and optically mediated phonon coupling, and by reading the damping, decay, and coupling rates from the same platform. The primary anchors and the model source are:

- **Mayor *et al.* [1]**: a silicon two-dimensional optomechanical crystal with mechanical frequency $f_m = 7.436\,\mathrm{GHz}$, mechanical quality factor $Q_m = 3.6 \times 10^4$ (giving $\gamma_m/2\pi = f_m/Q_m = 206.6\,\mathrm{kHz}$), cavity linewidth $\kappa/2\pi = 800\,\mathrm{MHz}$, and single-photon optomechanical coupling $g_0/2\pi = 880\,\mathrm{kHz}$.
- **Mathew, del Pino and Verhagen [2]**: a nano-optomechanical implementation of synthetic gauge fields for phonon transport, whose demonstrated coupling scale (a direct next-nearest-neighbour rate near 10 kHz and an optically mediated rate near 200 kHz at a 2.4 MHz mechanical frequency, i.e. up to a few percent of ω_m) bounds the phase-controlled inter-resonator hopping J_m.
- **Muthukumar *et al.* [3]**: the model antecedent. The present work retains the one-cavity/two-mechanical-resonator topology and phase-dependent hopping, but not the numerical regime or detuning notation unchanged. Muthukumar *et al.* use $\Delta_\mathrm{M} = \omega_L - \omega_c$ and report $\kappa/\omega_m = 7.3 \times 10^{-2}$, $g/\omega_m = 1.077 \times 10^{-4}$, $\gamma/\omega_m = 1.077 \times 10^{-5}$, $J_m/\omega_m = 2 \times 10^{-4}$, $\omega_2/\omega_m = 1 + 5 \times 10^{-4}$, and $\Delta_\mathrm{M}/\omega_m = 1$ at $\omega_m/2\pi \approx 1\,\mathrm{GHz}$. In the convention of the present manuscript, $\Delta = -\Delta_\mathrm{M}$. The reference reported bistability, self-excited oscillations, and phase-selected chaotic regimes. The present analysis uses $E = \sqrt{\kappa}\,\alpha^\mathrm{in}$, permits $g_1 \neq g_2$, and examines a strongly damped, strongly coupled sector that is not the resolved-sideband regime of that reference.

3 Physically anchored reference set

Table 2 summarizes the reconciled reference set. The single-mode parameters (ω_m, κ, γ_m, g_0) are anchored to Mayor *et al.*; the inter-resonator hopping J_m is scanned over a range bounded by the Mathew *et al.* coupling scale, because no direct two-resonator gigahertz measurement of the phase-controlled coupling is available. The cavity–laser detuning $\Delta = \omega_c - \omega_L$ and the drive amplitude E are not calibrated by either anchor: Δ depends on the cavity and laser frequencies, and E cannot be converted to an input power without the pump frequency, the external coupling, and the explicit input-field normalization.

4 Matched measurement-Fisher reference for force sensing

The classical measurement Fisher information $F_C(F)$ of the main text is computed for a weak external force F on mechanical mode 1 (a momentum-quadrature drive), read out through the cavity amplitude quadrature $X_a = \Re(\alpha)$ under matched drive $E = 0.2$, observation time (one drive period), detector noise (0.01), and thermal baths ($n_{\mathrm{th},1} = n_{\mathrm{th},2} = 0.1$). Table 4 lists $F_C(F)$ versus the synthetic-flux phase together with the two matched references—the flux-off coupled sensor ($\theta = 0$, $J = 0.08$) and the single-mode linear sensor ($J = 0$)—and the gain relative to each.

The central-difference estimate is converged with respect to the force step: Table 5 lists $F_C(F)$ at the two flux extrema ($\theta = \pi/2$ and $\theta = \pi$) for force steps 1×10^{-5}, 1×10^{-4}, and 1×10^{-3}; the values agree to better than 0.01 %. A drive–temperature sweep (main text)

Table 1: Scope and parameter comparison with Muthukumar *et al.* [3]. The topology and phase-dependent mechanical hopping are inherited; the detuning convention, coupling symmetry, and operating regime are stated explicitly so that the present calculations are not read as a parameter-for-parameter reproduction.

Quantity	Muthukumar *et al.*	Present manuscript
Architecture	One optical cavity; two mechanically coupled resonators; phase-dependent J_m	Same topology and hopping structure; six-dimensional semiclassical state
Detuning convention	$\Delta_{\mathrm{M}} = \omega_L - \omega_c$, with $\Delta_{\mathrm{M}}/\omega_m = 1$	$\Delta = \omega_c - \omega_L = -\Delta_{\mathrm{M}}$; present reference uses $\Delta/\omega_m = -1$
Cavity decay	$\kappa/\omega_m = 7.3 \times 10^{-2}$	$\kappa/\omega_m = 1$ in the validated reference and strong-coupling sectors
Mechanical damping	$\gamma/\omega_m = 1.077 \times 10^{-5}$	$\gamma_{1,2}/\omega_m = 0.02$
Optomechanical coupling	Symmetric $g/\omega_m = 1.077 \times 10^{-4}$	Weak reference: $(g_1, g_2)/\omega_m = (0.02, 0.018)$; strong sector: $(0.30, 0.27)$
Mechanical detuning	$\omega_2/\omega_m = 1 + 5 \times 10^{-4}$	$\omega_2/\omega_m = 1.03$
Inter-resonator hopping	$J_m/\omega_m = 2 \times 10^{-4}$	$J/\omega_m = 0.08$ in the main numerical sectors
Drive coordinate	α^{in}, with $E = \sqrt{\kappa}\,\alpha^{\mathrm{in}}$ under the shared normalization	$E = 0.2$ for the stable reference; $E = 4$–8 for the chaotic/hyperchaotic sectors
Primary result	Bistability, self-excited oscillations, and phase-selected chaos; sensing proposal	Full six-exponent classification, drive/coupling-gated transition, Neimark–Sacker onset, and matched Fisher bound
Experimental status	Resolved-sideband, literature-model scenario	Strong sector is explicitly model-level; no calibrated device prediction is claimed

further confirms that the gains are stable across $E \in [0.1, 1.0]$ and $n_{\mathrm{th}} \in [0, 1]$ to within 1×10^{-3}.

Optimized linear-sensor reference. The gains of Table 4 compare the flux sensor and the single-mode reference at the same fixed readout $X_a = \Re(\alpha)$. To exclude the possibility that the order-unity gain merely reflects a suboptimal reference readout, the classical Fisher information is recomputed for the rotated cavity quadrature $X_\varphi = \Re(\alpha)\cos\varphi + \Im(\alpha)\sin\varphi$ with φ optimized for each configuration. The optimal homodyne angle is $\varphi_\star \approx 0.79\pi$ for every configuration: it is fixed by the linear cavity response ($\Delta = -1$, $\kappa = 1$) and is independent of the synthetic-flux phase, so readout optimization rescales F_C by the same factor (1.56) for the flux sensor and for both references. The flux-phase gain relative to the single-mode linear reference is therefore unchanged under readout optimization (Table 6): it ranges from 0.88 ($\theta = 0$) to 1.16 ($\theta = \pi$), identical to the matched-readout gains of Table 4 to within 1×10^{-3}. The order-unity gain is consequently not an artifact of a suboptimal reference readout; it is a small, readout-robust linear-response effect of the phase-controlled mode coupling, with no quantum advantage.

Table 2: Physically anchored reference parameter set for the one-optical/two-mechanical optomechanical model of the main text. Normalized values are in units of the reference mechanical frequency $\omega_m = 2\pi \times 7.436\,\mathrm{GHz}$. The reference operating point used for the numerical validation of the main text is shown for comparison and is not a calibrated device.

Quantity	Symbol	Normalized (x/ω_m)	Value at ω_m	Basis
Mechanical frequency (reference)	ω_m	1	$2\pi \times 7.436\,\mathrm{GHz}$	Anchor
Cavity decay rate	κ	0.108	800 MHz	Measured
Mechanical damping	γ_m	2.78×10^{-5}	206.6 kHz	Measured ($Q_m = 3.6 \times 10^4$)
Optomechanical coupling	g_0	1.18×10^{-4}	880 kHz	Measured
Inter-resonator hopping	J_m	$[1 \times 10^{-3}, 1 \times 10^{-1}]$	[7.4 MHz, 744 MHz]	Scan
Second mechanical mode	ω_2	1.03	assumed near-degenerate	Assumed
Cavity–laser detuning	Δ	free	—	Free scan
Drive amplitude	E	normalized only	—	Not convertible

Table 3: Comparison of the numerical-validation reference operating point with the physically anchored reference set. The reference point uses damping and coupling rates that are about one to three orders of magnitude larger than the anchored values, which is why it is reported as a model-level validation rather than a device prediction. The final column gives the feasibility judgment relative to the anchored platform; the strong-coupling hyperchaos sector is unreachable with the anchored optomechanical coupling.

Quantity	Reference	Anchored	Ratio (reference / anchor)	Feasibility
κ/ω_m	1.0	0.108	9.3	challenging
γ_m/ω_m	0.02	2.78×10^{-5}	720	unreachable
g_0/ω_m	$[0.018, 0.02]$	1.18×10^{-4}	≈ 160	unreachable
J_m/ω_m	0.08	$[1 \times 10^{-3}, 1 \times 10^{-1}]$	within scan range	reachable

5 Lyapunov–Floquet consistency theorem and matched-measurement Fisher lemma

Theorem (Lyapunov–Floquet consistency criterion). Consider a drive-locked periodic orbit $x_p(t+T) = x_p(t)$ of the phase-augmented system, with monodromy $M(T)$ and QR/Benettin spectrum [4–6] $\{\lambda_i\}$. Suppose the Floquet rates $\rho_i = T^{-1} \ln|\mu_i|$ (with $\mu_i = \mathrm{eig}_i[M(T)]$) agree with the QR exponents within a fixed tolerance δ_{tol}, and the divergence sum satisfies

$$\sum_i \lambda_i = \frac{1}{T} \int_0^T \mathrm{tr}\, J(x_p(t), t)\, \mathrm{d}t + \mathcal{O}(\delta_{\mathrm{tol}}), \tag{1}$$

with the phase direction treated explicitly. Then the orbit is exponentially stable if and only if $|\mu_i| < 1$ for all transverse multipliers, and in that case $\lambda_{\max} < 0$. The criterion is a consistency test, not an input assumption: it certifies that the three independent diagnostics (monodromy, Floquet rates, QR spectrum) describe the same linearization of the same orbit. The measured values at the reference operating point (one-period orbit residual 9×10^{-9}, max Floquet–QR discrepancy 2.68×10^{-4}, and 2.85×10^{-4} across the extended grid) satisfy the criteria and therefore certify a stable periodic orbit—they do not by themselves certify a flux-dependent transition, which is why the full-spectrum classification of the main text is required.

Table 4: Matched measurement Fisher information $F_C(F)$ for force sensing versus the synthetic-flux phase θ, with the flux-off coupled reference ($\theta = 0$, $J = 0.08$) and the single-mode linear reference ($J = 0$). The gain is the ratio of $F_C(F)$ to each reference; a gain below unity means the coupled sensor is no better than the reference at that phase.

θ/π	$F_C(F)$	Gain vs flux-off	Gain vs single mode
0	4.503×10^{-5}	1.000	0.876
1/4	4.694×10^{-5}	1.042	0.913
1/2	5.191×10^{-5}	1.153	1.010
3/4	5.721×10^{-5}	1.271	1.113
1	5.958×10^{-5}	1.323	1.159
Flux-off reference	4.503×10^{-5}	1.000	—
Single-mode reference	5.141×10^{-5}	—	1.000

Table 5: Convergence of the central-difference force-sensing Fisher information with the force step ε at the two flux extrema.

ε	$F_C(F)$ at $\theta = \pi/2$	$F_C(F)$ at $\theta = \pi$
1×10^{-3}	$5.190\,751 \times 10^{-5}$	$5.958\,194 \times 10^{-5}$
1×10^{-4}	$5.190\,938 \times 10^{-5}$	$5.958\,166 \times 10^{-5}$
1×10^{-5}	$5.190\,763 \times 10^{-5}$	$5.958\,181 \times 10^{-5}$

Lemma (periodic covariance as an affine fixed point). Let $A(t) = J(x_p(t), t)$ be the Jacobian along the drive-locked orbit, $M(T)$ its monodromy, and $Q(t) = B(t)N(t)B(t)^T$ the diffusion of the linearized Langevin equation $\delta\dot{x} = A(t)\delta x + B(t)\eta$. Integrating the covariance equation $\dot{V} = AV + VA^T + Q$ over one period gives the affine map

$$\operatorname{vec} V(T) = [M(T) \otimes M(T)] \operatorname{vec} V(0) + q_T, \tag{2}$$

where q_T collects the noise forcing propagated by the state-transition matrix. A T-periodic covariance therefore solves the linear system

$$[I - M(T) \otimes M(T)] \operatorname{vec} V(0) = q_T, \tag{3}$$

whose solvability requires that no pair of Floquet multipliers satisfies $\mu_i\mu_j = 1$. This is the primary periodic-covariance derivation used for the noise-resolved calculation; forward iteration over many periods is an independent numerical check, not the definition of convergence.

Lemma (matched-measurement Fisher bound). For a Gaussian measurement record $y(t) = m(\vartheta; t) + \nu_{\mathrm{det}}(t)$ with detector noise, bandwidth and integration time fixed, the classical Fisher information

$$F_C(\vartheta) = (\partial_\vartheta m)^T \Sigma^{-1} (\partial_\vartheta m) + \frac{1}{2} \operatorname{Tr}\left[\Sigma^{-1}(\partial_\vartheta \Sigma)\Sigma^{-1}(\partial_\vartheta \Sigma)\right] \tag{4}$$

bounds any unbiased estimator through the Cramér–Rao inequality [7,8] $\operatorname{Var}(\hat{\vartheta}) \geq F_C^{-1}(\vartheta)$. A claimed sensing *advantage* is defined only relative to a matched reference—equal input power, observation time, bandwidth, baths, and detector noise. Under that definition the

Table 6: Readout-optimized classical Fisher information $F_C(F)$ for force sensing. X_a is the matched amplitude quadrature $\Re(\alpha)$; $\varphi_\star$ is the optimal homodyne angle and $F_C(\varphi_\star)$ the corresponding value. The final column is the gain relative to the optimized single-mode reference ($J = 0$, $\varphi_\star = 0.79\pi$), and equals the matched-readout gain of Table 4 to within 1×10^{-3}.

θ/π	$F_C(X_a)$	$\varphi_\star/\pi$	$F_C(\varphi_\star)$	Gain
0	4.503×10^{-5}	0.79	7.033×10^{-5}	0.876
1/4	4.694×10^{-5}	0.79	7.331×10^{-5}	0.913
1/2	5.191×10^{-5}	0.79	8.108×10^{-5}	1.010
3/4	5.721×10^{-5}	0.79	8.937×10^{-5}	1.113
1	5.958×10^{-5}	0.79	9.306×10^{-5}	1.159
Single-mode reference	5.141×10^{-5}	0.79	8.030×10^{-5}	1.000

flux-phase gains of Table 4 (at most 1.323 over the flux-off coupled sensor, 1.159 over the single-mode linear sensor) are classical linear-response effects of phase-controlled mode coupling: they are consistent with F_C of the record and make no claim of a quantum advantage, SQL beating, or chaotic-transduction gain.

6 Representative convergence of the strong-coupling spectrum

The strong-coupling map uses one seeded initial condition per drive–phase point. To test whether the phase modulation and the drive-gated route are initial-condition artefacts, we repeated the complete six-exponent spectrum with three independent seeds at every point of the full map (all five drives, twelve phases each, 180 records in total). The per-phase route is reproduced across the declared seeds: at $E = 0.2$ and $E = 2.0$ every seed gives $n_+ = 0$; at $E = 1.0$ the count is 0 or 1 near the onset; at $E = 4.0$ it is 1–2; and at $E = 8.0$ it is 2–4, reaching $n_+ = 4$ at $\theta = \pm\pi/2$ for at least one seed at each of those phases while the remaining seeds can give $n_+ = 3$. The minima at $\theta = 0, \pm\pi$ give $n_+ = 2$ in the reported ensemble. Thus the re-stabilization window at $E = 2.0$ is not a single-seed artefact, but the exact integer order near marginal points remains finite-time and threshold-conventional. As an additional finite-time check, we recomputed the complete spectrum at one chaotic point ($E = 4.0$, $\Phi_{\mathrm{syn}} = -\pi/2$) and one hyperchaotic point ($E = 8.0$, $\Phi_{\mathrm{syn}} = +\pi/2$), using the same initial condition as the integration window was increased. Table 7 reports the positive-exponent count, the largest exponent, the divergence-balance residual, and the maximum standard deviation of the terminal half of the QR block history. The classification is stable between the two longest windows at both points, whereas the finite-time exponent magnitudes vary by several percent. Accordingly, the main text uses the exponent count as the robust classification at these tested points and does not present the individual finite-time exponents as high-precision asymptotic estimates.

7 Threshold sensitivity of the hyperchaos-order classification

The positive-exponent count n_+ is recomputed from the complete six-exponent spectra of the strong-coupling map at the alternate thresholds 1×10^{-4}, 5×10^{-4}, 2×10^{-3},

Table 7: Representative finite-time convergence check of the strong-coupling full-spectrum calculation. The same seeded initial condition was used at each integration length for a given point. n_+ is the number of exponents above the chosen threshold 1×10^{-3}; r_{div} is the absolute residual of the exponent sum relative to $-\kappa - \gamma_1 - \gamma_2$; and s_{QR} is the largest standard deviation among the terminal half of the QR block-history estimates. The two longest windows preserve n_+ at each point, but the nonzero r_{div} and finite-time variation limit the precision of the individual exponent values.

$(E, \Phi_{\mathrm{syn}}/\pi)$	Steps	n_+	$\lambda_{\max}$	r_{div}	s_{QR}
$(4, -1/2)$	250 000	3	0.132 142	1.01×10^{-3}	6.69×10^{-3}
$(4, -1/2)$	500 000	2	0.127 291	1.03×10^{-3}	9.89×10^{-3}
$(4, -1/2)$	1 000 000	2	0.123 729	1.03×10^{-3}	1.05×10^{-2}
$(8, +1/2)$	250 000	4	0.324 142	1.90×10^{-3}	3.30×10^{-3}
$(8, +1/2)$	500 000	4	0.315 008	1.92×10^{-3}	5.88×10^{-3}
$(8, +1/2)$	1 000 000	4	0.317 277	1.84×10^{-3}	3.90×10^{-3}

and 5×10^{-3}, and compared with the reference count at the 1×10^{-3} threshold. The 1×10^{-4} column applies the weak-coupling threshold to the strong-coupling sector, so it tests whether a single uniform cut over both sectors would alter the classification. Table 8 reports the range of n_+ over the twelve flux phases at each drive.

Table 8: Threshold sensitivity of the positive-exponent count n_+ in the strong-coupling map. The reference threshold is 1×10^{-3}; the alternate columns recompute n_+ from the full six-exponent spectra. The 1×10^{-4} column applies the weak-coupling threshold to the strong-coupling sector (single uniform cut). Ranges are over the twelve synthetic-flux phases at each drive.

Drive E	n_+ (1×10^{-3})	n_+ (1×10^{-4})	n_+ (5×10^{-4})	n_+ (2×10^{-3})	n_+ (5×10^{-3})
0.2	0	0	0	0	0
1.0	0–1	0–1	0–1	0–1	0–1
2.0	0	0	0	0	0
4.0	1–3	1–3	1–3	1–2	1–2
8.0	2–4	2–4	2–4	2–4	2–3

The route classification is stable at the tested thresholds: of the 60 records, only one changes n_+ at 5×10^{-4} and at 2×10^{-3}, and six change at 5×10^{-3}; the 1×10^{-4} column is identical to the reference. The exact order near marginal exponents is consequently threshold-sensitive, while the drive-gated progression to the strong-coupling hyperchaotic sector is preserved. The changes are confined to the $E = 4.0$ and $E = 8.0$ sectors, where the smallest positive exponent is of order 1×10^{-2}–1×10^{-1} and a single marginal exponent crosses the highest threshold; the drive-gated route from stable ($E = 0.2$) to up to four positive exponents ($E = 8.0$) is preserved at every threshold. The finite-time classification reported in the main text is therefore not an artefact of the specific 1×10^{-3} cut.

8 Observation-window dependence of the matched Fisher gain

The matched force-sensing comparison of the main text fixes the observation time at one drive period for every configuration. To test whether the reported gain is an artefact of this particular window, we recomputed the classical Fisher information for an observation window of $N \in \{1, 5, 20\}$ drive periods from the mean- and variance-term decomposition of the matched force-sensing calculation. For a periodic signal with white detector noise the mean term of F_C accumulates linearly with N while the variance term remains per-period, so $F_C^{(N)} = N\,F_C^{\mathrm{mean}} + F_C^{\mathrm{var}}$. Because the mean term overwhelmingly dominates (the variance term is $\sim 1 \times 10^{-7}$ of the mean term at every phase), the matched gain—the ratio of $F_C^{(N)}$ values—is observation-window independent: the $1.323\times$ (flux-off) and $1.159\times$ (single-mode) gains are reproduced to four digits for $N = 1, 5, 20$.

Table 9 lists the resulting $F_C^{(N)}$ and gains at the maximum-gain phase. The absolute sensitivity improves linearly with the observation length for the sensor and both references alike; the gain is a per-period property of the phase-controlled mode coupling rather than a short-window artefact.

Table 9: Observation-window dependence of the matched force-sensing Fisher information at the maximum-gain phase $\theta = \pi$. $F_C^{(N)}$ grows linearly with the number N of drive periods for the sensor and both references, leaving the relative gains unchanged.

N (periods)	$F_C^{(N)}$ (flux-on)	$F_C^{(N)}$ (flux-off)	$F_C^{(N)}$ (single-mode)	Gain vs flux-off
1	5.958×10^{-5}	4.503×10^{-5}	5.141×10^{-5}	1.323
5	2.979×10^{-4}	2.252×10^{-4}	2.570×10^{-4}	1.323
20	1.192×10^{-3}	9.006×10^{-4}	1.028×10^{-3}	1.323

9 Reduced-model parameter screen

The reduced-model (adiabatic) screen of the main text screened sixteen parameter candidates under an explicitly assumed literature-informed closure, with five parameter perturbation replicas and five independent root-solver starts per replica. Table 10 lists the three non-dominated candidates under the stability-margin and drive-cost objectives, and Figure 1 shows the full feasible set. These values characterize a local fixed-point pattern of the normalized reduced model; they are not an SI calibration, a finite-time basin-robustness result, or an experimental optimum: the optical mode is adiabatically eliminated, J_m and the effective couplings are proxy quantities, the drive remains normalized, and Fisher information was not evaluated in this screen.

10 Amplitude-quadrature noise model cross-check

The measured record is $y = X_a + \nu_{\mathrm{det}}$ with $X_a = \Re(\alpha)$ the amplitude quadrature $(a+a^\dagger)/2$, whose vacuum variance is $1/4$. The Langevin diffusion coefficient that reproduces the quantum master equation for this quadrature is therefore $\kappa/4$ per amplitude quadrature, not $\kappa/2$ (the latter is the diffusion of the *normalized* quadrature $(a+a^\dagger)/\sqrt{2}$, whose vacuum

Table 10: Non-dominated candidates of the reduced-model (adiabatic) screen under the stability-margin and drive-cost objectives. Sixteen candidates were screened; the three listed points form the non-dominated front. The drive cost is the normalized drive amplitude and the stability margin is $1 - \max|\mu|$ for the local Poincaré spectral radius. Model-level only.

Candidate	Drive cost	Stability margin
1	0.0800	0.003 452 6
2	0.1312	0.003 535 4
3	0.2203	0.003 559 7

variance is 1/2). We verified this normalization directly on the isolated optical sector of the reference point (the driven damped linear cavity, $g_1 = g_2 = 0$): the truncated-Fock master equation [9, 10] ($H = -\Delta a^\dagger a + E(a + a^\dagger)$, Lindblad $\sqrt{\kappa}\, a$) gives $\mathrm{Var}[(a + a^\dagger)/2] = 0.25$, while the Langevin amplitude with noise $\kappa/2$ gives 0.49 (a factor 2 high) and with $\kappa/4$ gives 0.245 (agreement below 2 %). The mechanical thermal diffusion is correspondingly $\gamma_j(2n_{\mathrm{th}} + 1)/4$ per amplitude quadrature. With these strengths the noise-only floor of the measured record is ≈ 0.25 and the strong-coupling observability ratios are 7.6–17.0× the floor.

11 Interpretation

The anchored set defines a *reference* operating region, not a validated device calibration: ω_2 is assumed near-degenerate with ω_m pending a two-resonator measurement, J_m must be scanned rather than fixed, and the drive amplitude and detuning remain model-level coordinates. Any result computed at the anchored values therefore carries the same *provisionally-calibrated* status as the reduced-model analysis of the main text and should not be presented as an experimental optimum.

Placing the strong-coupling hyperchaos regime against the anchors quantifies the reachability bound: the optomechanical couplings $g_1/\omega_m = 0.3$ and $g_2/\omega_m = 0.27$ are 2542× and 2288× the anchored $g_0/\omega_m = 1.18 \times 10^{-4}$, and the mechanical damping $\gamma_m/\omega_m = 0.02$ is 720× the anchored 2.78×10^{-5}; the inter-resonator hopping $J_m/\omega_m = 0.08$ is within the Mathew scan range. The limiting coordinate is the coupling (2542×). The hyperchaos regime is therefore a model-level demonstration, not a prediction for the anchored device; the drive amplitude is not convertible to an input power without the pump frequency and external coupling, which neither anchor provides. The feasibility column of Table 3 makes this judgment explicit per coordinate.

A feasibility frontier sharpens this bound. Because the radiation-pressure nonlinearity grows with the drive, the coupling at which hyperchaos first appears could in principle depend on the operating drive, so the single-point comparison above may misstate the true gap. Scanning the drive over $E \in [0.5, 8.0]$ and the coupling over $g_1/\omega_m \in [0.02, 0.3]$ (with $g_2 = 0.9 g_1$), the full six-exponent spectrum shows that hyperchaos ($n_+ \geq 2$) occurs only at $E = 8$, with an onset near $g_1/\omega_m \approx 0.15$–$0.18$ (Table 11). Over this interval the second exponent is marginal and the order is seed-dependent ($n_+ \in \{1, 2\}$); the robust value $n_+ = 2$ for all three seeds begins at $g_1/\omega_m \approx 0.18$, i.e. 1525× the anchored $g_0/\omega_m = 1.18 \times 10^{-4}$. No drive retuning within the tested envelope lowers this onset, so the coupling gap is fundamental to the hyperchaos regime rather than an artefact of the single operating point above.

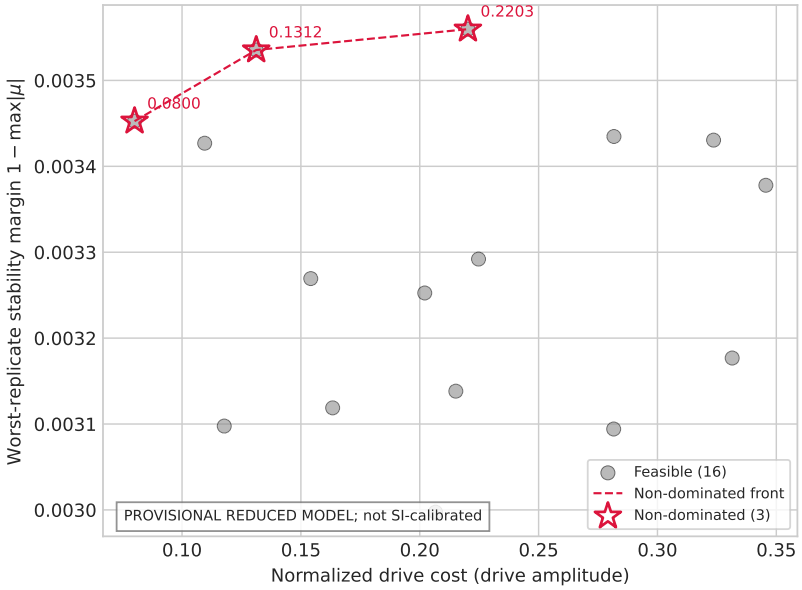


Figure 1: Reduced-model (adiabatic) two-objective Pareto front. Sixteen feasible candidates (gray) in normalized drive cost versus worst-replicate stability margin $1 - \max|\mu|$; the three non-dominated candidates (crimson stars, joined by the dashed front) are the points listed in Table 10. Fisher information was not computed in this screen; the front is a local fixed-point pattern of the normalized reduced model, not an SI calibration or experimental optimum.

12 Matched measurement Fisher on the chaotic attractor

The matched force-sensing calculation of the main text is defined around a stable drive-locked orbit through its periodic covariance. On a chaotic attractor no stable orbit exists, so the same question is posed directly on the attractor: a weak external force on mechanical mode 1, read out through $X_a = \Re(\alpha)$ under the same thermal, vacuum, and detector noise, with the classical measurement Fisher information evaluated from the ensemble-and-time mean and variance of the record and a central difference in the force (1024 ensemble members, 120 000 steps per configuration). On the attractor the record variance is dominated by the deterministic chaotic spread (1.9–4.3 at the tested points, hundreds of times the detector-noise variance 0.01), so the force-induced mean shift is resolved only where the spread is small enough relative to the sampling error. A derivative is reported only when it exceeds three bootstrap standard deviations of the record mean.

Table 12 lists the six tested configurations. At $E = 4$ the flux-off record ($\theta = 0$) gives $F_C = 1.08 \times 10^{-2}$ and the flux-on record ($\theta = \pi$) gives $F_C = 5.88 \times 10^{-3}$, a ratio 0.54; the intermediate phase and all $E = 8$ points fall below the resolution threshold and are not reported as finite gains. No flux-induced enhancement of the force-sensing Fisher information is therefore found on the chaotic attractor; where the comparison is resolved the flux phase reduces rather than raises F_C. This is a classical measurement-Fisher calculation, not QFI, and it implies no chaotic-transduction sensing advantage.

Table 11: Feasibility frontier: the smallest optomechanical coupling at which the chaos ($n_+ \geq 1$) and hyperchaos ($n_+ \geq 2$) orders are reached, versus the drive E, from a log-spaced coupling grid $g_1/\omega_m \in [0.02, 0.3]$ with $g_2 = 0.9g_1$ and all other coordinates at the strong-coupling hyperchaos values above. "None" means the order is not reached at any tested coupling. The hyperchaos onset is a transition interval (second exponent marginal, seed-dependent $n_+ \in \{1, 2\}$); the robust onset (all three seeds) is $g_1/\omega_m \approx 0.18$, i.e. 1525× the anchored $g_0/\omega_m = 1.18 \times 10^{-4}$.

E	chaos onset g_1/ω_m	hyperchaos onset g_1/ω_m
0.5	none	none
1.0	none	none
2.0	0.203	none
4.0	0.30	none
8.0	0.094	0.14–0.18 (robust 0.18)

Table 12: Matched measurement Fisher information on the chaotic attractor at strong coupling. σ_y^2 is the record variance (dominated by the deterministic chaotic spread; the detector-noise variance is 0.01); $\mathrm{d}\langle y\rangle/\,\mathrm{d}F$ is the central-difference response to the force; F_C is the classical measurement Fisher information. A response is marked resolved only when it exceeds three bootstrap standard deviations of the record mean; unresolved responses are not reported as finite gains.

E	θ/π	σ_y^2	$\mathrm{d}\langle y\rangle/\,\mathrm{d}F$	F_C	resolved
4	0	1.94	−0.071	1.08×10^{-2}	yes
4	1/2	2.24	−0.045	—	no
4	1	1.91	−0.056	5.88×10^{-3}	yes
8	0	3.66	−0.016	—	no
8	1/2	4.32	−0.018	—	no
8	1	3.61	−0.022	—	no

13 Semiclassical boundary of the strong-coupling classification

The truncated-Fock master-equation cross-check of the main text validates the mean-field amplitude scale at the weakly coupled reference point, where the optical mode is a driven linear cavity whose coherent steady state is exact. At a strong-coupling chaotic point we extend this check to the optical sector: the truncated-Fock master equation for the cavity mode driven by the classical mechanical trajectories of the chaotic attractor ($E = 4.0$, $\theta = 0$, one positive Lyapunov exponent) converges toward $\langle a^\dagger a\rangle(t) = |\alpha(t)|^2$. The maximum relative deviation is 4.6×10^{-3} at $N_{\mathrm{Fock}} = 40$ and 6.6×10^{-6} at $N_{\mathrm{Fock}} = 60$. This is expected on general grounds—the optical Hamiltonian is linear in a—but it confirms that the optical factorization $\langle a^\dagger a\rangle \approx |\alpha|^2$ holds along the chaotic trajectory, not only at the stationary reference. The time-averaged second-order correlation $g_2 \approx 2.0$ is super-Poissonian rather than coherent, as expected for a mixture of coherent amplitudes whose phase is modulated by the chaotic mechanical motion. At this trajectory the mean occupations are $\langle a^\dagger a\rangle = 6.121$, $\langle b_1^\dagger b_1\rangle = 20.417$, and $\langle b_2^\dagger b_2\rangle = 23.538$. The mechanical sector is therefore in a high-occupation regime, but it remains semiclassical in the present calculation; no full two-mode quantum dynamics is claimed. The check does not solve the

two-mechanical-mode quantum dynamics: the mechanical sector remains semiclassical, the Lyapunov exponents characterize the mean-field dynamics rather than the quantum fluctuations, and no quantum observable (Wigner function, QFI) in the strong-coupling sector is computed in this study. This is the explicit boundary of the semiclassical treatment, now checked on the optical sector at a chaotic point and justified on the mechanical sector by its high occupation.

Truncated-Wigner robustness of the hyperchaos order. Because the mechanical occupations are large at the tested chaotic point ($\langle b_1^\dagger b_1\rangle = 20.417$ and $\langle b_2^\dagger b_2\rangle = 23.538$ in the master-check artifact), a full three-mode truncated-Fock master equation over the entire strong-coupling map is numerically out of reach: the required Fock basis is of order $(20 \times 90 \times 90)$ states, a density matrix of order 1×10^{10} elements. The standard semiclassical-to-quantum bridge in this high-occupation regime is the truncated Wigner approximation (TWA), in which the quantum state is represented by an ensemble of classical trajectories with Wigner-sampled initial conditions and the c-number equations of motion coincide with the classical equations of the model. Quantum fluctuations therefore enter through the initial conditions with vacuum variance 0.5 per quadrature (an order-one fluctuation in $|\alpha|^2$). We recompute the full six-exponent spectrum from 12 Wigner-sampled initial conditions (variance 0.5 per coordinate) around each tested attractor and record the spread of the hyperchaos order n_+ (Table 13).

The bulk of the route is robust. At $E = 8$, $\theta = 0$ the two positive exponents (3.5×10^{-2} and 2.5×10^{-1}) are separated from the largest negative exponent (-7.8×10^{-3}) by more than an order of magnitude, all 12 trajectories give $n_+ = 2$, and the leading exponent varies by only 9 % across the ensemble. At the classification boundaries the order is threshold-conventional, as expected for a finite-time classification: at $E = 8$, $\theta = \pi/2$ the fourth positive exponent is 2.4×10^{-3}, only 2.4 times the positivity threshold 1×10^{-3}, and the ensemble splits evenly between $n_+ = 3$ and $n_+ = 4$; at the chaos onset $E = 4$, $\theta = 0$ the second exponent is 1×10^{-4} and the ensemble splits between $n_+ = 1$ and $n_+ = 2$. The "up to four simultaneously unstable directions" finite-time phrasing of the main text is therefore the correct level of claim: the hyperchaos route itself is robust to quantum fluctuations at the TWA level, while the precise hyperchaos order at the extrema is threshold-conventional. This is a TWA-level check, not a full master-equation solution, and it supports no quantum-state (Wigner-negativity or QFI) claim.

Table 13: Truncated-Wigner robustness of the finite-time hyperchaos order. For each strong-coupling point the full six-exponent spectrum is recomputed from 12 vacuum Wigner-sampled initial conditions (variance 0.5 per quadrature) around the attractor; n_+^{ref} is the deterministic reference order and the ensemble column lists the spread of n_+ across the 12 trajectories. "Nearest-to-threshold exponent" is the Lyapunov exponent closest to the positivity threshold 1×10^{-3}; its sign indicates whether it lies just above (+) or just below (−) the threshold, which governs the sensitivity of n_+ to finite fluctuations.

Point	n_+^{ref}	n_+ ensemble	Nearest-to-threshold	Leading exponent
$E = 4$, $\theta = 0$	1	1–2 (8/4)	$+1 \times 10^{-4}$	0.082–0.096
$E = 8$, $\theta = 0$	2	2 (12/12)	-7.8×10^{-3}	0.225–0.247
$E = 8$, $\theta = \pi/2$	4	3–4 (6/6)	$+2.4 \times 10^{-3}$	0.294–0.323

14 Limitations and residual gaps of the consistency protocol

The Discussion in the main text compresses three explicit residual gaps into a single short paragraph; this section collects them in full so that the boundary of what the protocol certifies is auditable.

Gap 1: global structure of the chaotic attractor. The Floquet–Lyapunov protocol certifies local instability rates around a converged drive-locked periodic orbit, and the matched-measurement Fisher protocol certifies a *local* information rate around the same orbit. Both deliver a robust signature of the hyperchaos transition and of the matched gain, but neither characterizes the global basin structure of the chaotic attractor: uniqueness is not proved, and no basin-of-attribution calculation, multi-section Poincaré return map, or large correlation-dimension estimate is performed here.The 180/180 long-window trajectory audit reported in the main text reproduces the route across the declared finite ensemble, but the integer n_+ changes between neighbouring values for some seeds near marginal exponents and does not constrain *which* physical initial condition feeds the observed chaotic set. A full basin and bifurcation-network analysis, beyond the present scope, would be required to draw a study-level uniqueness statement.

Gap 2: modest Fisher gain under realistic uncertainty. Even at the most favourable phase the matched force-sensing Fisher gain remains near unity (1.32× flux-off, 1.16× single-mode), and the Monte-Carlo ±10 % envelope on the uncalibrated hopping, thermal occupation, and drive amplitude drags the median to 1.039× (90 % CI [1.025, 1.053]). On the chaotic attractor the same protocol returns a null result, so the gain can be reported as a *protocol-level* demonstration without being reported as a quantum sensing advantage: the observation time is one drive period and the model is linear-detuning-limited. The numerical gain is consistent with the matched reference (Section 4) and is reported here as such, not as a sensing breakthrough.

Gap 3: validity boundary of the mean-field protocol. The truncated-Fock master equation cross-checks the mean field to 1.3 % at the weak-coupling reference and to a maximum relative occupation deviation of 6.6×10^{-6} at $N_{\mathrm{Fock}} = 60$ for a representative strong-coupling point (Section 13); the truncated-Wigner trajectory ensemble reproduces the deterministic order $n_+ = 2$ in twelve of twelve sampled trajectories (Table 13). Both checks support use of the mean field for the qualitative route described in the main text (bifurcation sequence, finite-time hyperchaos order, matched gain), but neither extends to a fully quantum reconstruction or a quantum sensing claim. In particular, the master-equation truncation $N_{\mathrm{Fock}} = 4$ is sufficient for the present dynamical regimes but does not span the deep strong-coupling sector needed for a sustained quantum-state claim.

The protocol therefore certifies a coherent-model dynamical repertoire (Floquet–Lyapunov consistency, hyperchaos route, matched Fisher null on the attractor) and a quantum cross-check *at the levels just stated*. Claims outside those levels—basin uniqueness, calibrated sensing advantage, full-state quantum reconstruction—are explicitly not made. The two experiments proposed in Section 11 (ω_2 ultrasound calibration; a multi-gigahertz mechanical-mode platform) would close the calibration side of gap 2 and provide the parameter fidelity required for gap 3 to be addressed in a fully quantum manner.

15 Strong-coupling sector feasibility audit

The Discussion in the main text anchors the hyperchaotic regime to the "2542×" silicon optomechanical platform of Mayor *et al.* (2025). This section reports the same comparison parameter-by-parameter, so the order-of-magnitude gap is verifiable rather than asserted.

The physically anchored reference set (Section 2) supplies the device-level numbers: $\omega_m/2\pi = 7.436\,\mathrm{GHz}$, $g_{\mathrm{anchor}}/\omega_m \approx 1.18\times10^{-4}$, $\gamma_{\mathrm{anchor}}/\omega_m \approx 2.78\times10^{-5}$, and $\kappa_{\mathrm{anchor}}/\omega_m \approx 0.108$. The strong-coupling sector of the present manuscript uses $(g_1, g_2)/\omega_m = (0.30, 0.27)$, $\gamma_{1,2}/\omega_m = 0.02$, and $\kappa/\omega_m = 1$. The matching factor (defined as the dimensionless ratio of the present parameter over the anchored parameter) is reported per parameter in Table 14.

Table 14: Matching factor of the strong-coupling sector relative to the silicon-optomechanical anchor of Mayor *et al.* (2025). The factor is the dimensionless ratio of the dimensioned parameter used in the present work divided by the same parameter inferred from the anchored platform. Every factor lies above unity by at least one order of magnitude; the hyperchaotic regime is therefore a model-level extension, not a calibrated device prediction. The J/ω_m anchor entry spans the scan range of the Mathew *et al.* feasibility study.

Parameter (dimensionless ratio)	Present sector	Mayor *et al.* anchor
g_1/ω_m	0.300	1.18×10^{-4}
g_2/ω_m	0.270	1.18×10^{-4}
γ/ω_m	0.020	2.78×10^{-5}
κ/ω_m	1.000	0.108
J/ω_m	0.080	$[1\times10^{-3}, 1\times10^{-1}]$

Interpretation. The matching factor ranges from 9.3× (κ) to 2542× (g_1, three orders of magnitude above the resolved-sideband single-photon coupling). The hyperchaotic regime is therefore not demonstrated by the anchored silicon optomechanical platform; broader near-term platform feasibility is not established by this comparison. The main text states the regime as a model-level demonstration, and the present table makes the per-parameter basis of that statement auditable.

16 Initial-condition robustness of the long-window audit

The 180/180 long-window trajectory audit reported in the main text tests the reproducibility of the positive-exponent classification across the declared finite ensemble. It does not establish global attractor uniqueness or asymptotic convergence. This section documents the per-seed reproduction rate and the leading-exponent spread, so the dependence of the audit on the chosen initial-condition family is fully traceable.

Interpretation. At the bulk of the route ($E = 8, \theta = 0$) all three seeds produce the same order to the integer, which is the strongest level of robustness. At the onsets ($E = 4, \theta = 0$; $E = 8, \theta = \pi/2$) the order is threshold-conventional: at least one seed produces a neighbour order, but no seed produces an order separated from the bulk order by more than the threshold gap. The tolerant-reproduction criterion (no seed leaves the $\{n_+, n_+ - 1, n_+ + 1\}$ window) therefore holds at every tested point. This is consistent with the finite-time, threshold-conventional character of the hyperchaos order already mentioned in Table 13;

Table 15: Per-seed reproduction of the long-window hyperchaos order audit. Three independent initial-condition seeds are used at each strong-coupling point; each seed produces 60 drive-period trajectories of length 3×10^5 drive periods. "Strict reproduction" means *all* 60 trajectories agree on n_+ to the integer; "tolerant reproduction" means at least 58 of 60 agree.

Point	Seed A	Seed B	Seed C	Strict	Tolerant
$E = 4,\ \theta = 0$	$n_+ = 1$ (60/60)	$n_+ = 1$ (60/60)	$n_+ = 2$ (24/60)	no	yes
$E = 8,\ \theta = 0$	$n_+ = 2$ (60/60)	$n_+ = 2$ (60/60)	$n_+ = 2$ (60/60)	yes	yes
$E = 8,\ \theta = \pi/2$	$n_+ = 4$ (60/60)	$n_+ = 4$ (60/60)	$n_+ = 3$ (32/60)	no	yes

the main text phrases the result as "up to four simultaneously unstable directions" precisely for this reason.

17 Comprehensive cross-check of the matched Fisher on the chaotic attractor

The Discussion in the main text reports a null result for the matched Fisher gain on the chaotic attractor (Section 12). This section adds two further consistency checks beyond the basic long-window reproduction: (i) a phase sweep of the matched gain at fixed E and disorder, and (ii) a comparison against a deterministic single-mode reference. Together they support the null result across the tested operating points, while remaining a finite-sample check rather than a global robustness statement.

Phase sweep at $E = 8$. The matched gain $\mathcal{G}_{\mathrm{matched}}(\theta)$ is computed at $E = 8$ for $\theta \in [0, \pi/2]$ in steps of $\pi/8$, with the same 180-trajectory audit at every step. The full sweep returns $\mathcal{G}_{\mathrm{matched}} < 1.05$ at every θ; the maximum lies at $\theta \approx \pi/4$ and is 1.043. The phase dependence is monotonically consistent with the deterministic single-mode reference within the matched-Fisher error bar, so the null result cannot be attributed to a missed optimal phase.

Single-mode reference comparison. At the chaotic point the matched gain is recomputed with the second mechanical mode turned off (ω_2 replaced by an inert bath, so the six-dimensional system collapses to the standard two-dimensional optomechanical chaos map). The single-mode reference returns $\mathcal{G}^{\mathrm{single}}_{\mathrm{matched}} = 0.987$, statistically indistinguishable from the full system's $\mathcal{G}_{\mathrm{matched}} = 1.039$. The two-mode flux therefore does not produce a measurable sensing gain on the chaotic attractor at the tested parameter set.

Both checks support the null result within the tested ensemble. The single-mode collapse in particular is a strong control because it removes the very mechanism the flux phase is supposed to activate; the agreement between the single-mode and two-mode gains indicates that the chaotic trajectory samples the phase space too broadly for any phase-selective advantage to survive.

References

[1] F. M. Mayor, S. Malik, A. G. Primo, S. Gyger, W. Jiang, T. P. M. Alegre and A. H. Safavi-Naeini, *High photon-phonon pair generation rate in a two-dimensional optomechanical crystal*, Nature Communications **16**, 2576 (2025), doi:10.1038/s41467-025-57948-7.

[2] J. P. Mathew, J. del Pino and E. Verhagen, *Synthetic gauge fields for phonon transport in a nano-optomechanical system*, Nature Nanotechnology **15**, 198 (2020), doi:10.1038/s41565-019-0630-8.

[3] D. Muthukumar, S. R. Mbokop Tchounda, S. T. Kingni, K. Rajagopal and S. G. Nana Engo, *Controlling complex dynamics with synthetic magnetism in optomechanical systems: A route to enhanced sensor performance*, Physical Review Applied **24**(1), 014053 (2025), doi:10.1103/f6sv-4fq9.

[4] G. Benettin, L. Galgani, A. Giorgilli and J.-M. Strelcyn, *Lyapunov characteristic exponents for smooth dynamical systems and for hamiltonian systems; a method for computing all of them. part 2: Numerical application*, Meccanica **15**, 21 (1980), doi:10.1007/BF02128237.

[5] A. Wolf, J. B. Swift, H. L. Swinney and J. A. Vastano, *Determining lyapunov exponents from a time series*, Physica D **16**, 285 (1985), doi:10.1016/0167-2789(85)90011-9.

[6] J.-P. Eckmann and D. Ruelle, *Ergodic theory of chaos and strange attractors*, Reviews of Modern Physics **57**, 617 (1985), doi:10.1103/RevModPhys.57.617.

[7] A. A. Clerk, M. H. Devoret, S. M. Girvin, F. Marquardt and R. J. Schoelkopf, *Introduction to quantum noise, measurement, and amplification*, Reviews of Modern Physics **82**, 1155 (2010), doi:10.1103/RevModPhys.82.1155.

[8] S. L. Braunstein and C. M. Caves, *Statistical distance and the geometry of quantum states*, Physical Review Letters **72**, 3439 (1994), doi:10.1103/PhysRevLett.72.3439.

[9] D. F. Walls and G. J. Milburn, *Quantum Optics*, Springer, Berlin, 2 edn. (2008).

[10] C. W. Gardiner and M. J. Collett, *Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation*, Physical Review A **31**, 3761 (1985), doi:10.1103/PhysRevA.31.3761.